

Set 7: Customer Demographics Analysis

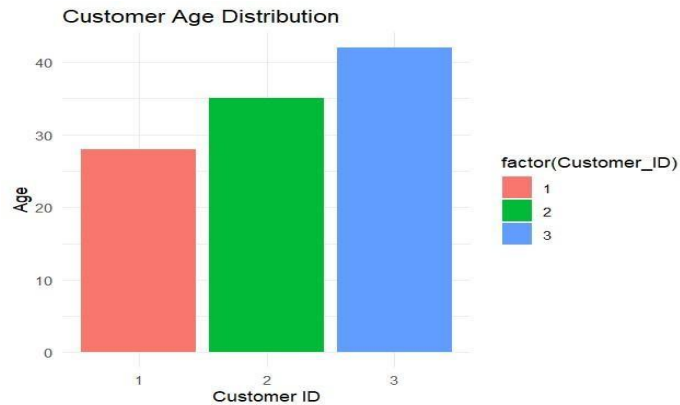
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
library(ggplot2)

customers <- data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

ggplot(customers, aes(x = factor(Customer_ID), y = Age, fill =
factor(Customer_ID))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age"
  ) +
  theme_minimal()
```

Output:



2. Generate a pie chart to represent the distribution of customers by gender.

```
library(ggplot2)
```

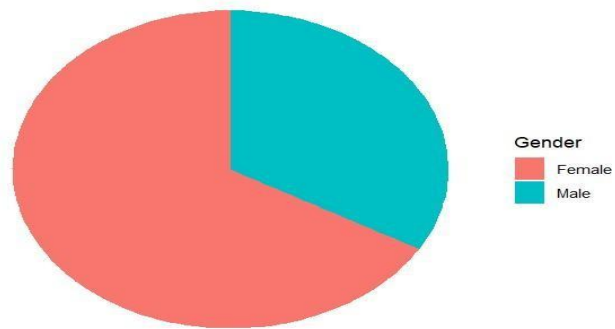
```
customers <- data.frame(
  Customer_ID = c(1,2,3),
  Gender = c("Female","Male","Female")
)
```

```
gender_count <- as.data.frame(table(customers$Gender))
colnames(gender_count) <- c("Gender","Count")
```

```
ggplot(gender_count, aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Customer Gender Distribution") +
  theme_void()
```

Output:

Customer Gender Distribution



3. Build a table to show the demographic information for each customer. Label the table elements.

```
customers <-  
data.frame(  
  Customer_ID =  
    c(1,2,3),  
  Age = c(28,35,42),  
  Gender =  
    c("Female","Male"  
      "Female"),  
  Income =  
    c(50000,60000,7500  
      0) )print(customers)
```

Output:

	Customer_ID	Gender
1	1	Female
2	2	Male
3	3	Female

Set 8: Employee Performance Analysis

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
library(ggplot2)
```

```
employee <- data.frame(
```

```
  Employee_ID = c(1, 2, 3),
```

```
  Department = c("Sales", "HR", "Marketing"),
```

```
  Years_of_Service = c(5, 3, 7),
```

```
  Performance_Score = c(85, 92, 78)
```

```
)
```

```
ggplot(employee, aes(x = Years_of_Service, y = Performance_Score, group = 1))  
+
```

```
  geom_line(color = "blue", linewidth = 1) +
```

```
  geom_point(color = "red", size = 3) +
```

```
  labs(
```

```
    title = "Employee Performance Trend",
```

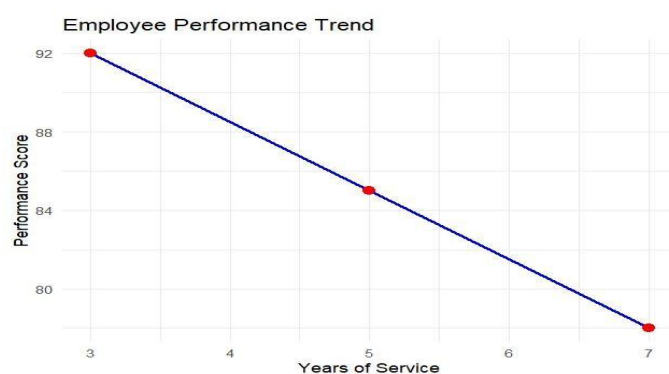
```
    x = "Years of Service",
```

```
    y = "Performance Score"
```

```
  ) +
```

```
  theme_minimal()
```

Output:



2. Generate a bar chart showing the distribution of employees across different

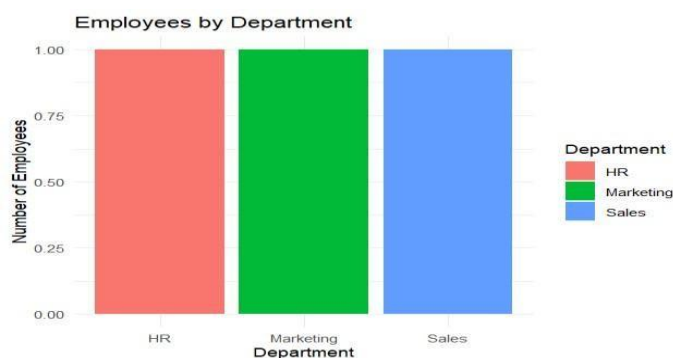
departments. Label the chart elements.

```
library(ggplot2)
```

```
employee <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)
```

```
ggplot(employee, aes(x = Department, fill = Department)) +  
  geom_bar() +  
  labs(  
    title = "Employees by Department",  
    x = "Department",  
    y = "Number of Employees"  
  ) +  
  theme_minimal()
```

Output:



3. Build a table to display the performance data for each employee. Label the

table elements.

Program:

```
employee <-  
data.frame(  
  Employee_ID =  
c(1, 2, 3),  
  Department =  
c("Sales", "HR",  
  "Marketing"),  
  Years_of_Service =  
c(5, 3, 7),  
  Performance_Score  
= c(85, 92, 78)  
)
```

```
print(employee)
```

Output:

	Employee_ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by

Course. Add title and labels. Interpret student performance.

```
library(ggplot2)
```

```
learning <- data.frame(
```

```
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
```

```
  Gender = c("Male","Female","Male","Female","Male","Female"),
```

```
  Age = c(20,22,19,21,23,20),
```

```
  Course = c("R","R","SQL","R","R","SQL"),
```

```
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
```

```
  Videos_Watched = c(12,15,8,18,9,14),
```

```
  Quiz_Score = c(78,85,65,92,70,88),
```

```
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
```

```
    "2025-02-08","2025-03-12","2025-03-12")
```

```
)
```

```
ggplot(learning, aes(x = Quiz_Score)) +
```

```
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
```

```
  labs(
```

```
    title = "Distribution of Quiz Scores",
```

```
    x = "Quiz Score",
```

```
    y = "Frequency"
```

```
) +
```

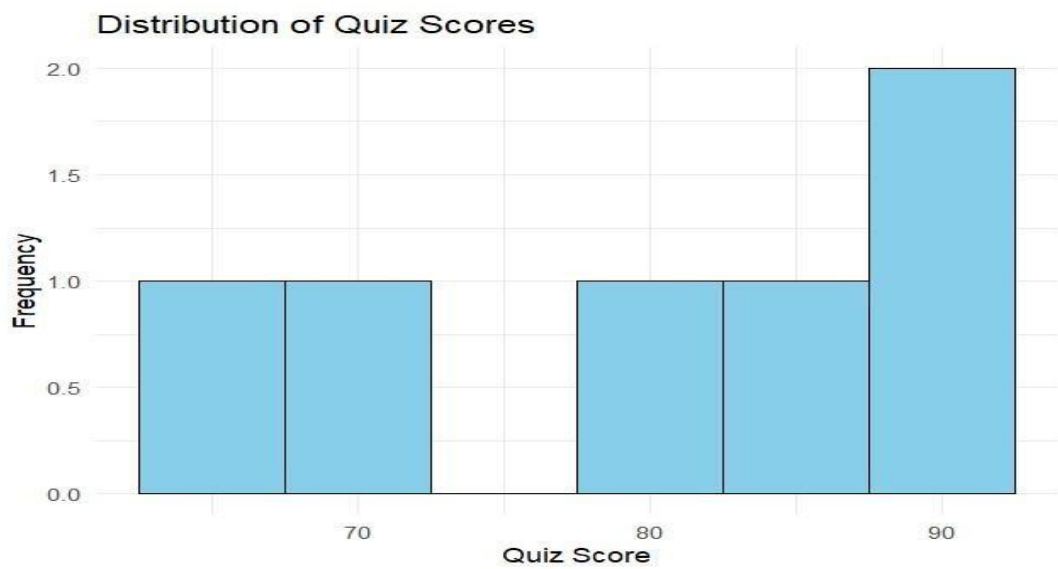
```
  theme_minimal()
```

```
ggplot(learning, aes(x = Course, y = Quiz_Score, fill = Course)) +
```

```
  geom_boxplot() +
```

```
labs(  
  title = "Quiz Scores by Course",  
  x = "Course",  
  y = "Quiz Score"  
) +  
theme_minimal()
```

Output:



2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent

Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

```
library(ggplot2)
```

```
learning <- data.frame(
```

```
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
```

```
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
```

```
  Quiz_Score = c(78,85,65,92,70,88),
```

```
  Videos_Watched = c(12,15,8,18,9,14)
```

```
)
```

```
ggplot(learning, aes(x = Study_Time, y = Quiz_Score, size =  
Videos_Watched)) +
```

```
  geom_point(alpha = 0.6, color = "blue") +
```

```
  labs(
```

```
    title = "Study Time vs Quiz Score",
```

```
    x = "Study Time (Hours)",
```

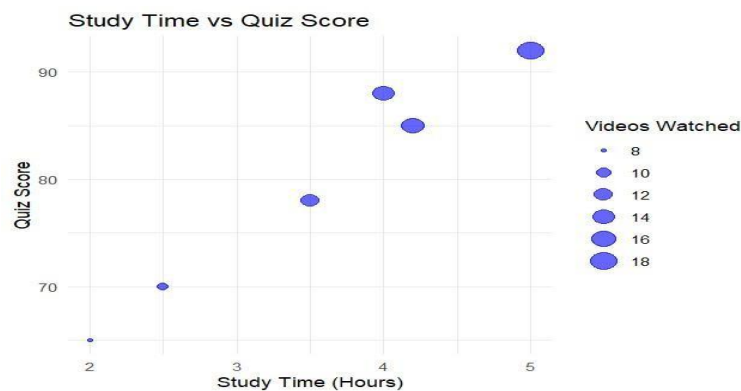
```
    y = "Quiz Score",
```

```
    size = "Videos Watched"
```

```
) +
```

```
  theme_minimal()
```

Output:



3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

```
library(ggplot2)
```

```
library(dplyr)
```

```
learning <- data.frame(  
  Student_ID = c("L01","L02","L03","L04","L05","L06"),  
  Quiz_Score = c(78,85,65,92,70,88),  
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",  
    "2025-02-08","2025-03-12","2025-03-12")  
)
```

```
learning$Login_Date <- as.Date(learning$Login_Date)
```

```
learning$Month <- format(learning$Login_Date, "%Y-%m")
```

```
monthly_avg <- learning %>%
```

```
  group_by(Month) %>%
```

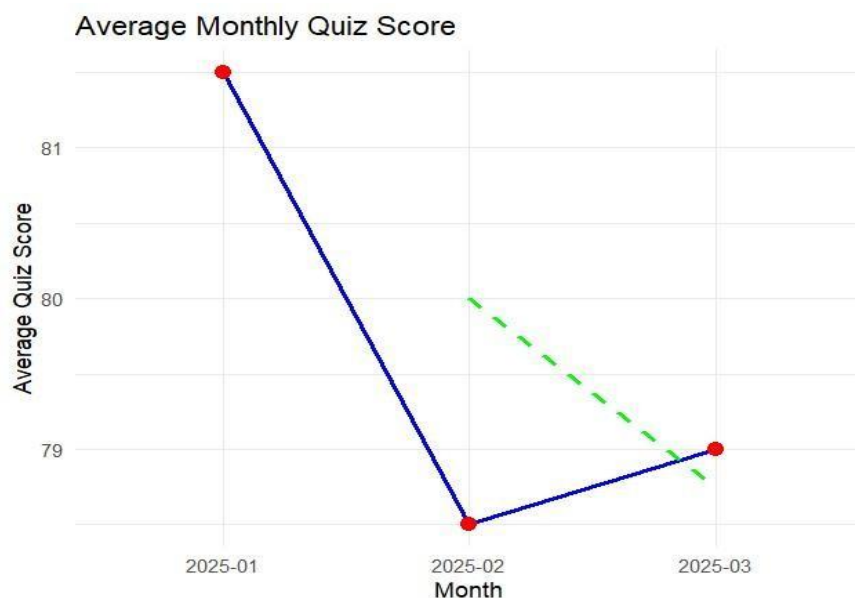
```
  summarise(Average_Score = mean(Quiz_Score))
```

```

monthly_avg$Moving_Average <- stats::filter(
  monthly_avg$Average_Score,
  rep(1/2, 2),
  sides = 1
)

ggplot(monthly_avg, aes(x = Month, y = Average_Score, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(size = 3, color = "red") +
  geom_line(aes(y = Moving_Average),
    color = "green",
    linewidth = 1,
    linetype = "dashed") +
  labs(
    title = "Average Monthly Quiz Score",
    x = "Month",
    y = "Average Quiz Score"
  ) +
  theme_minimal()
Output:

```



Set 10: Survey Responses Analysis

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

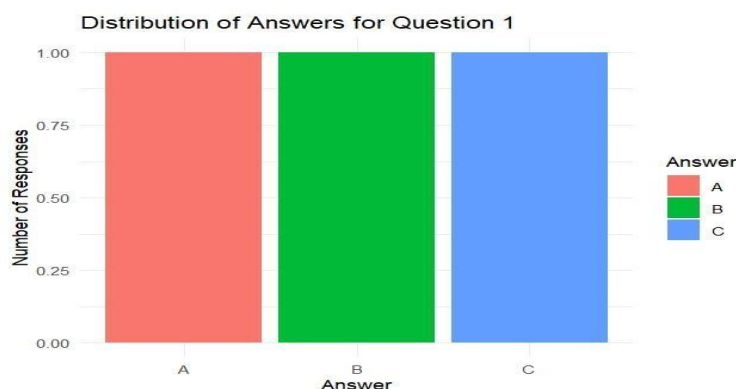
```
library(ggplot2)
```

```
survey <- data.frame(  
  Survey_ID = c(1, 2, 3),  
  Question1 = c("A", "B", "C"),  
  Question2 = c("B", "A", "A"),  
  Question3 = c("C", "D", "B")  
)
```

```
q1 <- as.data.frame(table(survey$Question1))  
colnames(q1) <- c("Answer", "Count")
```

```
ggplot(q1, aes(x = Answer, y = Count, fill = Answer)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Distribution of Answers for Question 1",  
    x = "Answer",  
    y = "Number of Responses"  
  ) +  
  theme_minimal()
```

Output:



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
library(ggplot2)
```

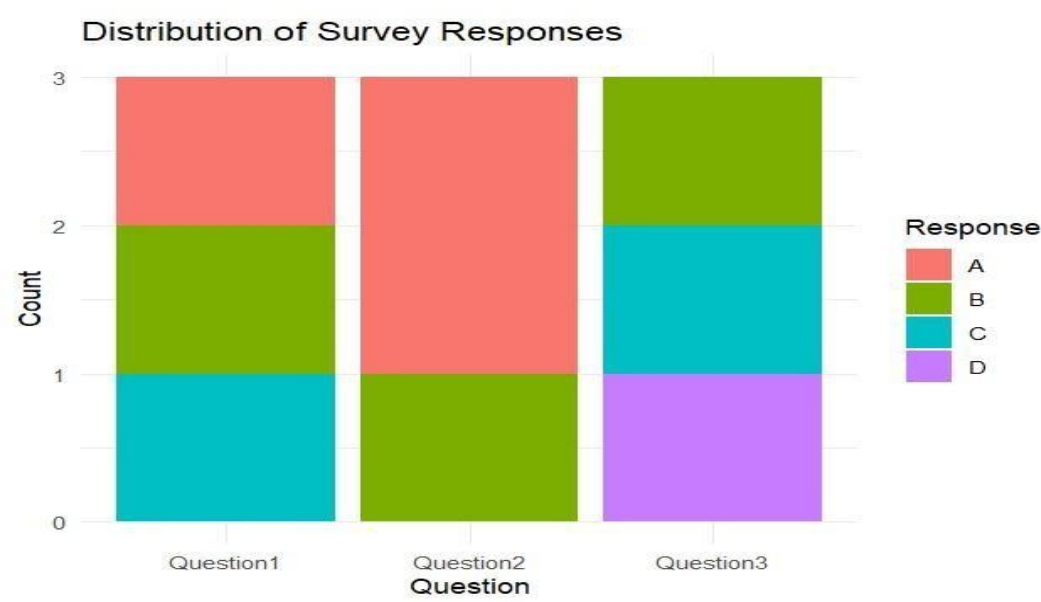
```
library(tidyr)
```

```
survey <- data.frame(  
  Survey_ID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

```
survey_long <- pivot_longer(  
  survey,  
  cols = c(Question1, Question2, Question3),  
  names_to = "Question",  
  values_to = "Response"  
)
```

```
ggplot(survey_long, aes(x = Question, fill = Response)) +  
  geom_bar() +  
  labs(  
    title = "Distribution of Survey Responses",  
    x = "Question",  
    y = "Count"  
  ) +  
  theme_minimal()
```

Output:



3. Build a table to show the survey response data for each survey. Label the table elements.

Program:

```
survey <-  
data.frame(  
  Survey_ID =  
c(1,2,3),  
  Question1 =  
c("A","B","C"  
"),  
  Question2 =  
c("B","A","A"  
"),  
  Question3 =  
c("C","D","B"  
")  
)
```

```
print(survey)
```

Output:

	Survey_ID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

Set 11: Product Category Analysis

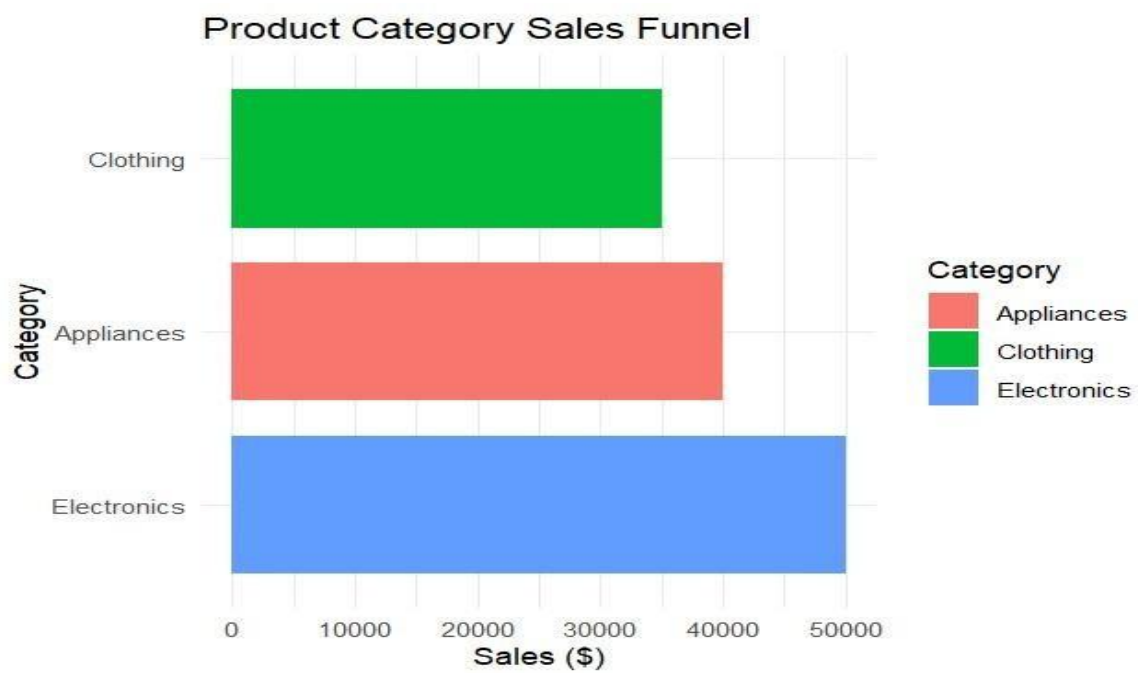
1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
library(ggplot2)
```

```
category <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)
```

```
ggplot(category, aes(x = reorder(Category, -Sales), y = Sales, fill = Category)) +  
  geom_bar(stat = "identity", width = 0.8) +  
  coord_flip() +  
  labs(  
    title = "Product Category Sales Funnel",  
    x = "Category",  
    y = "Sales ($)"  
  ) +  
  theme_minimal()
```


Output:



2. Build a table to display the sales data for each product category. Label the table elements.

```
category <- data.frame(
  Category = c("Electronics","Clothing","Appliances"),
  Sales = c(50000,35000,40000)
)

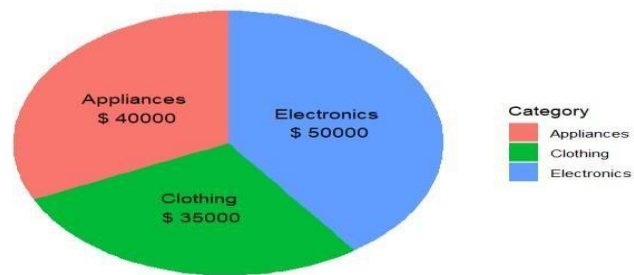
print(category)

library(ggplot2)

ggplot(category, aes(x = "", y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  geom_text(
    aes(label = paste(Category, "\n$", Sales)),
    position = position_stack(vjust = 0.5),
    size = 4
  ) +
  labs(
    title = "Distribution of Sales Across Product Categories"
  ) +
  theme_void()
```

Output:

Distribution of Sales Across Product Categories



Set 12: Dataset: Website Traffic

```
library(ggplot2)
```

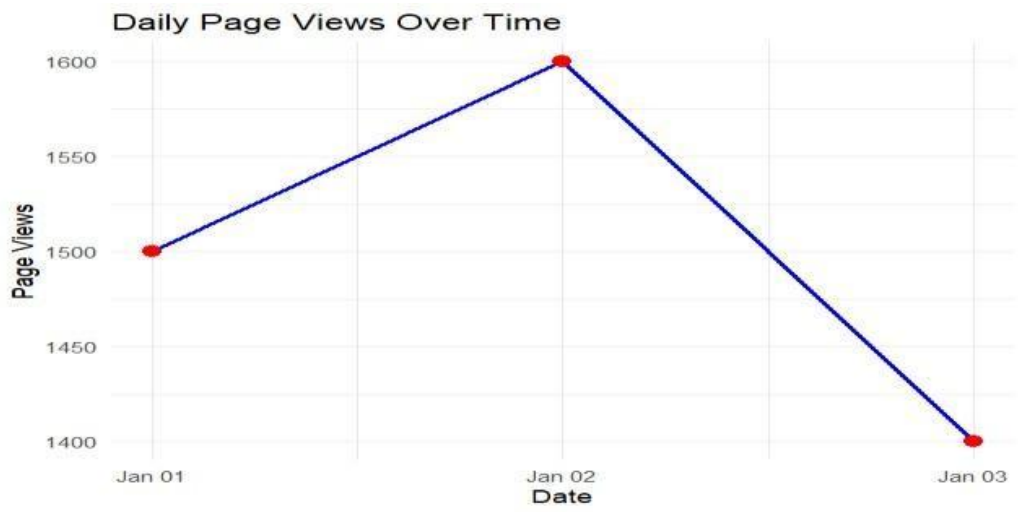
```
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01",  
                    "2023-01-02",  
                    "2023-01-03")),  
  Page_Views = c(1500,1600,1400),  
  Click_Through_Rate = c(2.3,2.7,2.0)  
)
```

```
ggplot(traffic, aes(x = Date, y = Page_Views, group = 1)) +  
  geom_line(color = "blue", linewidth = 1) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Daily Page Views Over Time",  
    x = "Date",  
    y = "Page Views"  
  ) +
```

theme_minimal()

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

Output:



1. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

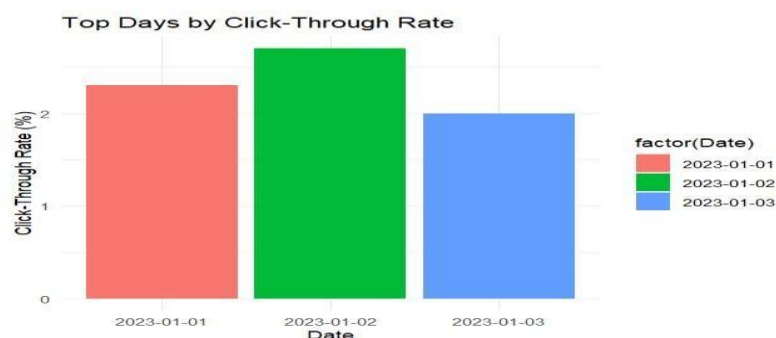
```
library(ggplot2)
```

```
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01",  
    "2023-01-02",  
    "2023-01-03")),  
  Click_Through_Rate = c(2.3,2.7,2.0)  
)
```

```
top_days <- traffic[order(-traffic$Click_Through_Rate), ]
```

```
ggplot(top_days, aes(x = factor(Date), y = Click_Through_Rate, fill =  
  factor(Date))) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Top Days by Click-Through Rate",  
    x = "Date",  
    y = "Click-Through Rate (%)"  
  ) +  
  theme_minimal()
```

Output:



2. Build a table to show the website traffic data for each day. Label the table elements.

```
traffic <-  
data.frame(  
  Date =  
c("2023-01-  
01", "2023-01-  
02", "2023-01-  
03"),  
  Page_Views  
=  
c(1500,1600,14  
00),  
  Click_Throug  
h_Rate =  
c("2.3%","2.7  
%","2.0%")  
)
```

```
print(traffic)
```

Output:

	Date	Click_Through_Rate
1	2023-01-01	2.3
2	2023-01-02	2.7
3	2023-01-03	2.0